

Smart Health – Change Summary and Extended Features

1. Project Overview

Smart Health started as a basic appointment booking platform and evolved into a more complete digital healthcare management system. The latest version adds patient-facing portals, hospital-admin controls, main-admin management, digital record handling, telemedicine support, and real-time communication features.

2. What Changed from the Previous Version

Previous State

- Basic appointment booking was available.
- Patients could browse centers and doctors.
- Notifications and queue tracking existed but were limited.
- Admin workflows were partial or disconnected.
- Telemedicine and feedback flows were incomplete.
- Real-time communication was not consistently validated.

New State

- Added patient portal workflows covering dashboard, appointments, reports, prescriptions, feedback, telemedicine, and health records.
- Improved role-based access between Hospital Admin and Main Admin.
- Added more complete administration, queue visibility, and feedback handling.
- Improved real-time consultation support using Socket.IO and WebRTC.
- Strengthened backend validation and database schema startup logic.
- Improved deployment compatibility for CORS, HTTPS, socket connections, and remote access.

3. Extended Features

Patient Features

- User dashboard and portal navigation
- Multi-step appointment booking with reason, symptoms, priority, and consultation type
- Telemedicine appointment support
- User feedback submission with hospital and doctor selection
- Digital health records and patient timeline
- Health reports upload, management, and download
- E-prescriptions access and viewing
- AI health assistant support for explanations and guidance
- Appointment status tracking and queue visibility

Admin Features

- Hospital admin role and scope-aware access
- Main admin portal for platform-wide management
- Admin dashboard metrics for appointments and queue status
- User management for admin creation, role updates, search, and deletion
- Feedback review and response workflows
- Telemedicine room access and consultation controls
- Report and prescription generation from appointment workflows

Real-Time Features

- Live queue updates through Socket.IO
- User notification updates
- Live consultation signaling using WebRTC + Socket.IO
- Chat support within telemedicine sessions
- Participant join/leave tracking during live sessions

4. New Frontend Routes

- /health-reports — upload and manage health reports
- /digital-health-records — view medical timeline and records
- /prescriptions — view, print, download, and ask AI about prescriptions
- /telemedicine — access the consultation room interface
- /ai-assistant — educational questions about medical records
- /hospital-admin/* — hospital-scoped admin portal
- /main-admin/* — platform-wide admin portal

5. Workflow Enhancements

- Patients can now book appointments with consultation type selection, including telemedicine.
- Users can attach existing reports and prescriptions to a new appointment.
- Hospital admins can generate medical reports directly from appointments.
- Generated reports appear in patient records, dashboard, and appointment details.
- E-prescriptions are issued from appointment actions and linked to patient records.
- Patients can join telemedicine sessions from the appointment status page or telemedicine portal.

6. Recent Fixes and Improvements

- Fixed feedback form data saving for hospital and doctor mappings.
- Fixed schema logic to detect missing fields and create required database tables reliably.
- Fixed telemedicine consultation type handling.
- Improved room setup and media access stability for audio/video communication.

- Enabled secure camera and microphone handling for browser-based consultation flows.
- Improved CORS and deployment readiness for frontend/backend communication.
- Resolved stale backend port conflicts and validated the production build state.

7. Technology Stack

Frontend

- React
- Vite
- Tailwind CSS
- Axios
- React Router
- Socket.IO Client
- React Hot Toast

Backend

- Node.js
- Express.js
- Socket.IO
- JWT Authentication
- bcryptjs
- MySQL2
- Nodemailer

Database

- MySQL
- Structured models for users, centers, doctors, appointments, feedback, reports, prescriptions, and admin scope

8. Deployment Readiness

The application has been hardened for deployment by improving:

- backend CORS handling
- frontend host binding
- socket configuration for external access
- secure HTTPS requirements for microphone and camera access

Recommended deployment flow:

- Frontend: deploy on HTTPS-enabled hosting
- Backend: deploy on a server with proper environment variables and CORS settings
- Database: host MySQL in a stable environment
- Environment: configure JWT secrets, database credentials, and SMTP settings

9. Summary

The Smart Health project is now much closer to a real healthcare workflow platform than a basic demo. It combines appointment booking, queue tracking, digital records, report and prescription management, telemedicine, AI assistance, and admin operations in a role-aware system designed for practical use and future production rollout.

10. Final Outcome

This release extends the system beyond a simple appointment app into a broader healthcare experience for patients, administrators, and clinical workflows. The project now includes a stronger patient journey, better admin controls, and real-time digital consultation features that make it more production-ready and feature-complete.